

Cricket World Cup Dataset

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Dataset Import & Setup

Python Code (Run in IDE):

```
import pandas as pd
import numpy as np

# Load the Cricket World Cup Dataset
df = pd.read_csv('cricket_wc.csv')

# Preview the data
print(df.head())
print('Shape:', df.shape)
print('Columns:', df.columns.tolist())
```

Output:

```
   Player  Team  Matches  Runs  Wickets  Catches  Strike_Rate  Economy_Rate ...
0  India_P1  India      11   485         9         8         80.00         5.73 ...
1  India_P2  India       6   393        18        10         93.35         9.22 ...
2  India_P3  India      15   560         5         5        157.38         6.27 ...

Shape: (150, 15)
Columns: ['Player', 'Team', 'Matches', 'Runs', 'Wickets', ...]
```

Problem 1: Total Runs Scored Across All Players

Problem Statement: Find the total number of runs scored by all players across all World Cup editions.

Python Code (IDE Screenshot Simulation):

```
import pandas as pd
```

```
import numpy as np

df = pd.read_csv('Cricket_wc.csv')

# Problem 1: Total runs scored across all players
total_runs = df['Runs'].sum()
print('Total Runs Scored:', total_runs)
```

Output:

```
Total Runs Scored: 66101
```

Problem 2: Team with Highest Total Runs

Problem Statement: Identify which team accumulated the highest total runs across all players.

Python Code (IDE Screenshot Simulation):

```
# Problem 2: Team with highest total runs
team_runs = df.groupby('Team')['Runs'].sum()
best_team = team_runs.idxmax()
print('Team with Most Runs:', best_team)
print(team_runs.sort_values(ascending=False))
```

Output:

```
Team with Most Runs: Sri Lanka
Team
Sri Lanka      8217
India          8210
West Indies    8295
...
```

Problem 3: Average Batting Average of All Players

Problem Statement: Compute the mean batting average across all players in the dataset.

Python Code (IDE Screenshot Simulation):

```
# Problem 3: Mean batting average of all players
mean_batting_avg = df['Batting_Avg'].mean().round(2)
print('Mean Batting Average:', mean_batting_avg)
```

Output:

```
Mean Batting Average: 39.98
```

Problem 4: Player with Most Wickets

Problem Statement: Find the player who took the most wickets in the entire dataset.

Python Code (IDE Screenshot Simulation):

```
# Problem 4: Player with most wickets
top_wicket_taker = df.loc[df['Wickets'].idxmax(), 'Player']
max_wickets = df['Wickets'].max()
print('Top Wicket Taker:', top_wicket_taker)
print('Wickets Taken:', max_wickets)
```

Output:

```
Top Wicket Taker: Pakistan_P4
Wickets Taken: 25
```

Problem 5: Statistical Summary of Runs

Problem Statement: Generate descriptive statistics (min, max, mean, std, quartiles) for runs scored.

Python Code (IDE Screenshot Simulation):

```
# Problem 5: Descriptive statistics of Runs
runs_stats = df['Runs'].describe()
print(runs_stats)
```

Output:

```
count    150.000000
mean     440.673333
std       217.110478
min        51.000000
25%       262.500000
50%       476.000000
75%       616.000000
max       788.000000
```

Problem 6: Players with 2 or More Centuries

Problem Statement: Filter and display all players who have scored 2 or more centuries.

Python Code (IDE Screenshot Simulation):

```
# Problem 6: Players with >= 2 centuries
centurions = df[df['Centuries'] >= 2][['Player', 'Team', 'Centuries']]
print(centurions.to_string(index=False))
```

Output:

```
      Player      Team  Centuries
India_P3      India          2
India_P6      India          2
Australia_P3  Australia          2
England_P6     England          2
Pakistan_P3    Pakistan          2
... (47 players total)
```

Problem 7: Total Wickets per Team (Sorted)

Problem Statement: Calculate total wickets taken by each team and sort them in descending order.

Python Code (IDE Screenshot Simulation):

```
# Problem 7: Total wickets per team sorted descending
team_wickets = df.groupby('Team')['Wickets'].sum().sort_values(ascending=False)
print(team_wickets)
```

Output:

Team	
India	213
West Indies	198
Sri Lanka	197
England	188
Pakistan	188
New Zealand	169
Australia	158
Bangladesh	155
Afghanistan	140
South Africa	133

Problem 8: Players with Above-Average Batting Average

Problem Statement: Count the number of players whose batting average is above the dataset mean.

Python Code (IDE Screenshot Simulation):

```
# Problem 8: Count players above mean batting average
mean_avg = df['Batting_Avg'].mean()
above_avg_count = df[df['Batting_Avg'] > mean_avg].shape[0]
print('Players above mean batting avg:', above_avg_count)
```

Output:

Players above mean batting avg: 78

Problem 9: Total Runs Scored by Year (Edition)

Problem Statement: Group the data by World Cup Year and find total runs for each edition.

Python Code (IDE Screenshot Simulation):

```
# Problem 9: Total runs per World Cup year
runs_by_year = df.groupby('Year')['Runs'].sum()
print(runs_by_year)
```

Output:

Year	
2003	9854
2007	11168
2011	13823
2015	10526
2019	12381
2023	8349

Problem 10: Distribution of Player Roles

Problem Statement: Find how many players belong to each role (Batsman, Bowler, All-rounder, Wicket-keeper).

Python Code (IDE Screenshot Simulation):

```
# Problem 10: Count of players per role
role_distribution = df['Role'].value_counts()
print(role_distribution)
```

Output:

Role	
Bowler	47
All-rounder	38
Batsman	36
Wicket-keeper	29

Problem 11: Top 5 Players by Strike Rate

Problem Statement: Find the top 5 players with the highest batting strike rates.

Python Code (IDE Screenshot Simulation):

```
# Problem 11: Top 5 players by Strike Rate
top_sr = df.sort_values('Strike_Rate', ascending=False).head(5)
print(top_sr[['Player', 'Team', 'Strike_Rate']].to_string(index=False))
```

Output:

Player	Team	Strike_Rate
South Africa_P11	South Africa	158.24
Pakistan_P14	Pakistan	158.05
India_P3	India	157.38
New Zealand_P14	New Zealand	156.11
Sri Lanka_P11	Sri Lanka	155.73

Problem 12: Correlation Between Runs and Batting Average

Problem Statement: Calculate the Pearson correlation coefficient between Runs and Batting Average.

Python Code (IDE Screenshot Simulation):

```
# Problem 12: Correlation between Runs and Batting Average
correlation = df['Runs'].corr(df['Batting_Avg']).round(4)
print('Correlation (Runs vs Batting Avg):', correlation)
```

Output:

Correlation (Runs vs Batting Avg): 0.0128

Problem 13: Teams with Most Economical Bowlers (Economy < 6)

Problem Statement: For each team, count players with economy rate below 6.0 (economical bowlers).

Python Code (IDE Screenshot Simulation):

```
# Problem 13: Economical bowlers (Economy < 6) per team
eco_bowlers = df[df['Economy_Rate'] < 6.0].groupby('Team').size()
print(eco_bowlers)
```

Output:

Team	
Afghanistan	7
Australia	3
Bangladesh	5
England	5
India	8
New Zealand	3
Pakistan	6
South Africa	8
Sri Lanka	2
West Indies	2

Problem 14: Total Catches by Team

Problem Statement: Calculate and rank teams by total number of catches taken by their players.

Python Code (IDE Screenshot Simulation):

```
# Problem 14: Total catches per team (ranked)
team_catches = df.groupby('Team')[['Catches']].sum()
team_catches = team_catches.sort_values('Catches', ascending=False)
print(team_catches)
```

Output:

Team	Catches
India	132
Bangladesh	116
England	113
New Zealand	113
West Indies	109
Pakistan	93
Sri Lanka	91
South Africa	84
Australia	80
Afghanistan	77

Problem 15: Pivot Table: Runs by Team and Year

Problem Statement: Create a pivot table showing total runs scored by each team per World Cup year.

Python Code (IDE Screenshot Simulation):

```
# Problem 15: Pivot table - Runs by Team x Year
pivot = df.pivot_table(
    values='Runs', index='Team',
    columns='Year', aggfunc='sum', fill_value=0
)
print(pivot)
```

Output:

Year	2003	2007	2011	2015	2019	2023
Team						
Afghanistan	306	1094	1176	1543	1349	1075
Australia	567	1375	1931	210	0	2388
Bangladesh	823	1233	1075	765	1162	1165
England	1239	2031	1704	1447	314	286
India	2141	1463	1508	596	810	696
...						

Problem 16: Players with Five-Wicket Hauls

Problem Statement: Filter players who have taken at least one five-wicket haul in the World Cup.

Python Code (IDE Screenshot Simulation):

```
# Problem 16: Players with five-wicket hauls
five_wkt = df[df['Five_Wickets'] > 0][['Player', 'Team', 'Five_Wickets']]
print(f'Total players with 5-wicket hauls: {len(five_wkt)}')
print(five_wkt.head(10).to_string(index=False))
```

Output:

```
Total players with 5-wicket hauls: 100
   Player  Team  Five_Wickets
India_P1  India           2
India_P5  India           1
India_P9  India           2
India_P10 India           2
India_P11 India           2
... (100 players total)
```

Problem 17: Average Batting Average by Player Role

Problem Statement: Group players by their Role and find the mean batting average for each role.

Python Code (IDE Screenshot Simulation):

```
# Problem 17: Mean batting average by player role
avg_by_role = df.groupby('Role')['Batting_Avg'].mean().round(2)
print(avg_by_role)
```

Output:

Role	
All-rounder	36.21
Batsman	37.81
Bowler	44.37

Wicket-keeper	40.50
---------------	-------

Problem 18: Top 10 Run Scorers in the Dataset

Problem Statement: Identify the top 10 highest run scorers across all teams and World Cup editions.

Python Code (IDE Screenshot Simulation):

```
# Problem 18: Top 10 run scorers
top_10 = df.nlargest(10, 'Runs')[['Player', 'Team', 'Runs']]
print(top_10.to_string(index=False))
```

Output:

Player	Team	Runs
West Indies_P13	West Indies	788
Pakistan_P8	Pakistan	781
South Africa_P4	South Africa	779
Australia_P15	Australia	776
Afghanistan_P7	Afghanistan	774
South Africa_P13	South Africa	773
Bangladesh_P14	Bangladesh	770
Afghanistan_P3	Afghanistan	765
New Zealand_P8	New Zealand	761
Australia_P4	Australia	760

Problem 19: Percentile Analysis of Runs (NumPy)

Problem Statement: Use NumPy to compute the 25th, 50th, and 75th percentile of runs scored.

Python Code (IDE Screenshot Simulation):

```
# Problem 19: Numpy percentile analysis of Runs
percentiles = np.percentile(df['Runs'], [25, 50, 75])
print('25th Percentile (Q1):', percentiles[0])
print('50th Percentile (Median):', percentiles[1])
print('75th Percentile (Q3):', percentiles[2])
```

Output:

```
25th Percentile (Q1): 262.5
50th Percentile (Median): 476.0
75th Percentile (Q3): 616.0
```

Problem 20: Total Matches Played per Team

Problem Statement: Sum the total number of matches played by each team and rank them.

► **Python Code (IDE Screenshot Simulation):**

```
# Problem 20: Total matches per team (ranked)
matches_per_team = df.groupby('Team')['Matches'].sum().sort_values(ascending=False)
```



```
print(matches_per_team)
```

Output:

Team	
India	203
Sri Lanka	188
Pakistan	186
Bangladesh	184
South Africa	182
Afghanistan	182
New Zealand	181
West Indies	177
Australia	176
England	175